Introduction to Programming
COMP101
2025-26

Lab-05
Supervised

Objectives:
Exercise-01 Nested While loop - debug

OR

Work Assignment-02

Exercise-01: Debug

A programmer wants to output the following menu and operations to show flow of control between nested while loops.

Main Menu
A: Option-A code
B: Option-B code
C: Option-C code
X: Exit

Enter an option A,B,C or X to exit: h
Invalid Option:
Enter an option A,B,C or X to exit: a

Option-A under development
Now back in outer loop - show menu again unless option = X

Main Menu
A: Option-A code
B: Option-B code
C: Option-C code
X: Exit

Enter an option A,B,C or X to exit: b

Option-B under development
Now back in outer loop - show menu again unless option = X

Main Menu
A: Option-A code
B: Option-B code
C: Option-C code
X: Exit

Enter an option A,B,C or X to exit: c

Option_C under development
Now back in outer loop - show menu again unless option = X

```
Main Menu
A: Option-A code
B: Option-B code
C: Option-C code
X: Exit

Enter an option A,B,C or X to exit: x

Option-X - menu ended
End of inner loop to accept an option
Now back in outer loop - show menu again unless option = X

Now out of outer loop
end of program
```

However, by accident all the code has been left-aligned and needs to be re-indented.

Debug the following so it produces the output as required above.
There are no other bugs in the code other than wrong indentation.
You can cut&paste it into Python.

Code to debug:

```
option = ' '
while option not in 'X':
print ("Main Menu")
print ("A: Option-A code")
print ("B: Option-B code")
print ("C: Option-C code")
print ("X: Exit")
print ()
option = input("Enter an option A,B,C or X to exit: ")
option = option.upper()
while option not in('A','B','C','X'):
print ('Invalid Option: ')
option = input("Enter an option A,B,C or X to exit: ")
option = option.upper()
print()
if option == "A":
   print ("Option-A under development")
elif option == "B":
   print ("Option-B under development")
elif option =="C":
   print ("Option_C under development")
else:
   print ("Option-X - menu ended")
   print ("End of inner loop to accept an option")

print ("Now back in outer loop - show menu agaian unless option = X")
print()

print ("Now out of outer loop")
print ("end of program")
```

Exercise-01: SOLUTION

```
option = ' '
while option not in 'X':
   print ("Main Menu")
   print ("A: Option-A code")
   print ("B: Option-B code")
   print ("C: Option-C code")
   print ("X: Exit")
   print ()
   option = input("Enter an option A,B,C or X to exit: ")
   option = option.upper()
   while option not in('A','B','C','X'):
      print ('Invalid Option: ')
      option = input("Enter an option A,B,C or X to exit: ")
      option = option.upper()
   print()
   if option == "A":
      print ("Option-A under development")
   elif option == "B":
      print ("Option-B under development")
   elif option =="C":
      print ("Option_C under development")
   else:
      print ("Option-X - menu ended")
      print ("End of inner loop to accept an option")

   print ("Now back in outer loop - show menu agaian unless option = X")
   print()

print ("Now out of outer loop")
print ("end of program")
```